

Civil Engineering

Apply Mathematics Principles

Assignment 1 - Questions Only

Instructions: Answer all questions. Show all working and use correct units.

Section A

1. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\text{Percentage} = \frac{\text{Part}}{\text{Whole}} \times 100\%$$

2. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$A = \frac{1}{2}(a + b)h$$

3. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\bar{x} = \frac{\sum x}{n}$$

4. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Section B

5. Calculate cement-sand ratio quantities for a mortar mix. **(10 marks)**
6. Find the area of a trapezoidal drainage section. **(10 marks)**
7. Compute mean and range from site-test readings. **(10 marks)**
8. Create one realistic civil engineering problem based on this topic and solve it fully. **(20 marks)**